Class 9 2022-23

MOTION

Ch 8 | Class 9th (Science)

Handwritten Notes & Imp Questions

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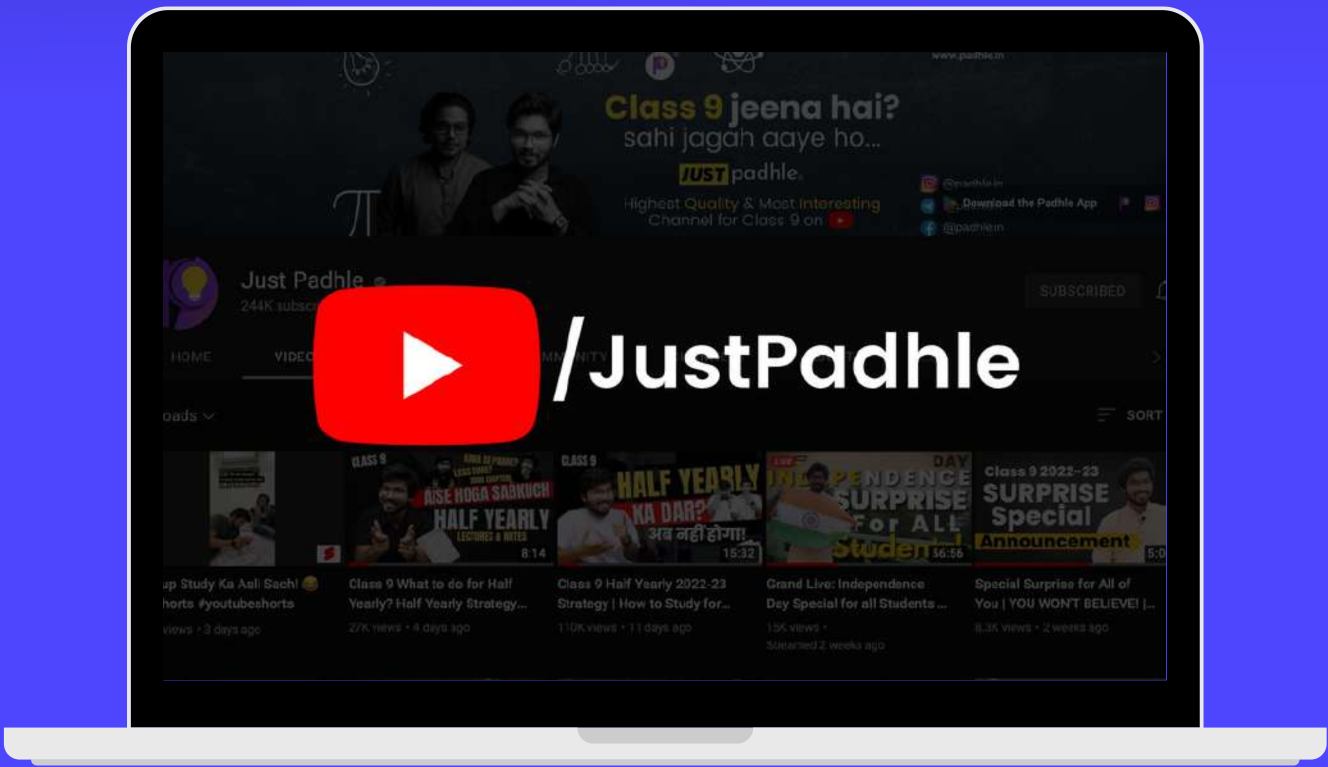

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Motion (जीवन कैसे चल रहा है)

* Definition

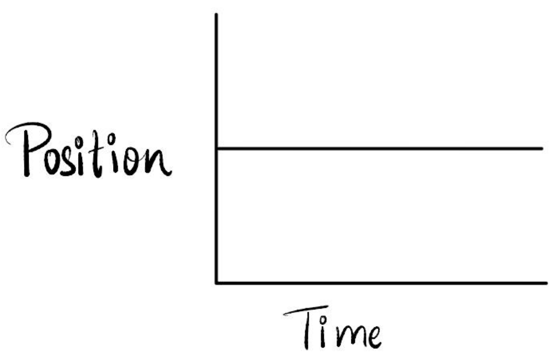
→ A body is said to be in motion when it changes its position with respect to time.
(जैसे जैसे time आगे बढ़ रहा है, वैसे वैसे position change हो रही है)

* Motion is relative with respect to observers.
(जैसे Gattu को teachers मस्तीखोर समझते है, मम्मी राजा बेटा समझती है और दोस्त बेवकूफ समझते है, means एक Gattu को अलग-अलग लोग, अलग-अलग नजरिये से देखते है, same goes with our Gattu, i.e. 'Motion')

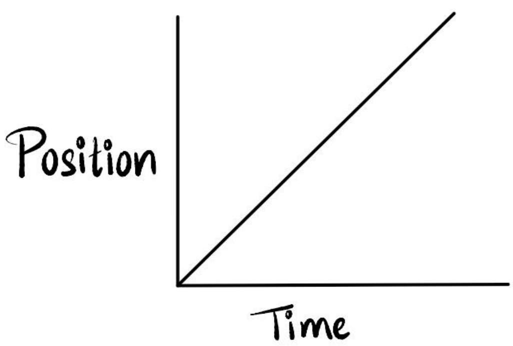
→ For a person moving in train, platform is in motion.
- But for a person standing on the platform, it is at rest.
- Platform is only one, but different observers view it differently.
#NazariyaOP.

* A body is said to be in —

i) Rest : → When it doesn't change position with respect to time.



ii) Motion : When it does change position with respect to time.



* Why with respect to time?

क्योंकि समय सबको एक नज़र से देखता है।

TYPES OF MOTION

1) Linear Motion

◊ Linear motion is a one-dimensional motion along a straight line.

2) Circulatory Motion

◊ The movement of a body following a circular path is called a circular motion.

3) Oscillatory Motion

◊ The movement of a body following to and fro path.

Distance V/s Displacement

Sr.	Distance	Displacement
1.	It is the length of the path travelled by a body while moving from initial to final position.	It is the shortest distance between the initial and the final position of the body.
2.	It is a scalar quantity.	It is a vector quantity.
3.	Distance measured is always positive.	Displacement can be negative or positive depending on reference point.
4.	The total distance covered = the algebraic sum of all the distance travelled in different directions.	The net displacement is the vector sum of the individual displacements in different directions.

5.	There is always a distance covered whenever there is a motion.	Displacement will be 0 if the body comes back to its initial position.
6.	Unit = metre (m)	Unit = metre (m)

Speed And Velocity

Sr.	Speed	Velocity
1.	It is the distance covered by the body per unit time.	It is the displacement of the body per unit time.
2.	Speed is a scalar quantity.	Velocity is a vector quantity.
3.	The average speed of a moving body can never be 0.	The average velocity of a moving body can be 0. (when displacement is 0)
4.	Speed gives an idea about rapidity of motion of body.	Velocity gives an idea about rapidity as well as position of body in motion.
5.	Speed of a moving body can never be negative.	Velocity of a moving body can be both positive, negative or 0.

a) Uniform Motion

→ A body is said to be in uniform motion when its velocity doesn't change.

b) Non-Uniform Motion

→ A body is said to be in non-uniform motion when its velocity changes with time.

Important Formulas

1) Average Speed

→ $\dfrac{\text{Total Distance Travelled}}{\text{Total time taken}}$ Example: $\dfrac{d_1 + d_2 + d_3}{t_1 + t_2 + t_3}$

2) Average Velocity

→ $\dfrac{\text{Sum of all velocities}}{\text{Number of velocities}}$ Examples: $\dfrac{V_1 + V_2 + V_3}{3}$

Acceleration

→ The rate of change of velocity with respect to time is called acceleration.

(Happens only in non-uniform motion क्योंकि उसी में Velocity change होती है
Otherwise in Uniform motion acceleration = 0, because velocity doesn't change.)

* Acceleration = $\dfrac{\text{final velocity} - \text{initial velocity}}{\text{time}}$ $\boxed{a = \dfrac{v-u}{t}}$

* Its SI unit is $\dfrac{\text{metre}}{\text{second}^2}$ $\left(\dfrac{m}{s^2}\right)$

- It is a vector quantity.
- The acceleration is taken to be positive if it is in the direction of velocity and negative when it is opposite to the direction of velocity.
- Negative acceleration is also named as retardation or deceleration.
- An object moving on a circular path though with uniform speed, is always said to be accelerated as it changes its direction every moment.

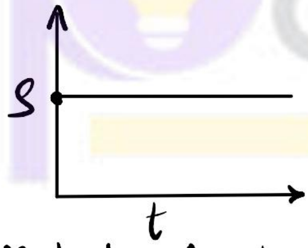
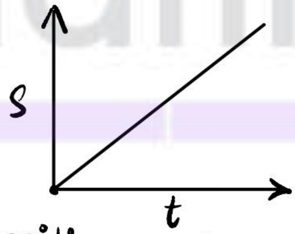
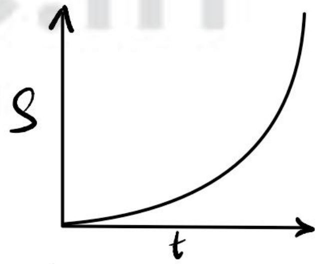
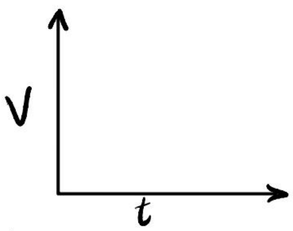
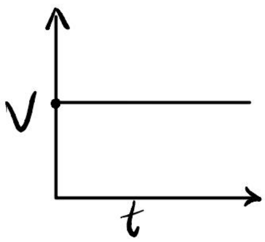
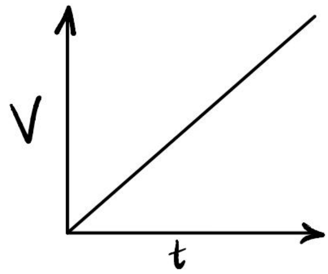
i) Uniform Acceleration

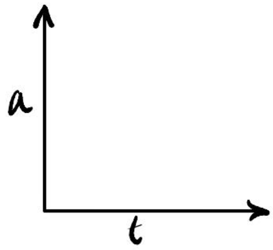
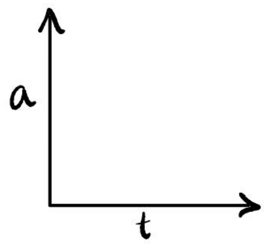
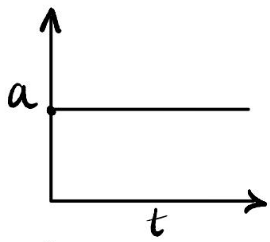
◊ When velocity of body changes by equal amounts in equal time intervals, acceleration is said to be uniform.
E.g. Motion of freely falling ball.

ii) Non-uniform Acceleration

◊ When velocity of body changes by unequal amounts in equal intervals of time, acceleration is said to be non-uniform.
E.g. Motion of car.

GRAPHICAL REPRESENTATION OF MOTION (Most important and easiest topic)

	For a body at rest	for a body in uniform motion	for a body in uniform acceleration (or non-uniform motion)
Displacement (or Distance) – Time Graph (s-t graph)	 ∵ body is at rest position doesn't change.	 position changes uniformly.	 Position changes exponentially.
Velocity-time graph (v-t graph)	 (0 graph because body is at rest, therefore v = 0)	 Velocity is constant	 Velocity changes uniformly

Acceleration-time graph (a-t graph)			
	 0 graph because body is at rest so $\underline{a=0}$	 0 graph because body has constant velocity so $\underline{a=0}$	 acceleration is uniform i.e. constant.

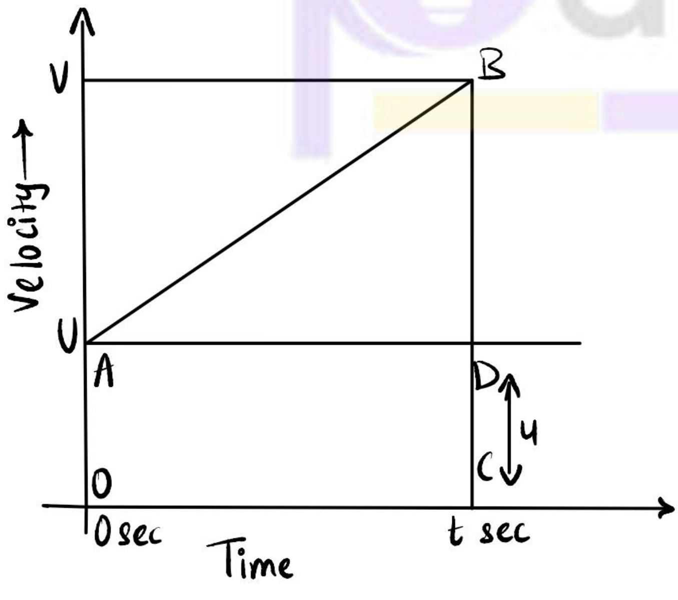
* Kinematic Equations (बहुत ही ज़्यादा important है, because numericals of this chapter + Gravitation depends on these)

i) $v = u + at$

ii) $s = ut + \frac{1}{2}at^2$

iii) $v^2 - u^2 = 2as$

* Derivation of these Kinematic Equations by Graphical Method (very important, usually asked directly in 5 marks).



* Second Equation : $s = ut + \frac{1}{2}at^2$

Distance travelled by object = Area of OBC (trapezium)

= Area of OADC (rectangle) + Area of ΔABD

$= OA \times AD + \frac{1}{2} \times AD \times BD$

$= u \times t + \frac{1}{2} \times t \times (v-u)$

$= ut + \frac{1}{2} \times t \times at$

$\Rightarrow s = ut + \frac{1}{2}at^2 \quad \left(\because a = \frac{(v-u)}{t}\right)$

Third Equation : $v^2 = u^2 + 2as$

s = Area of trapezium OABC

$s = \frac{(OA + BC) \times OC}{2}$

$s = \frac{(u+v) \times t}{2}$

$s = \left(\frac{u+v}{2}\right) \times \left(\frac{v-u}{2}\right)$

$s = \frac{v^2 - u^2}{2a}$

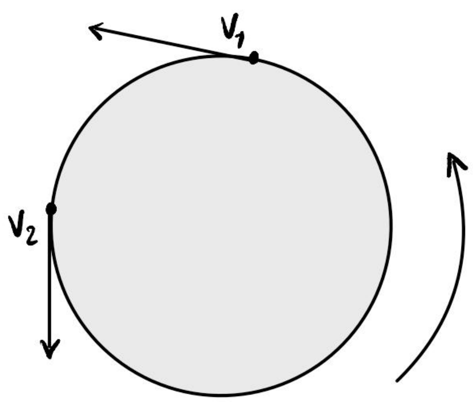
$\Rightarrow v^2 = u^2 + 2as$

Circular Motion

→ Object moving along circular path with uniform speed, (but not uniform velocity).

→ A body in circular motion can never have constant velocity, because at every instant it changes direction.

- Though the speed can be same, but since direction changes, velocity is not constant. (ये बात याद रखना)



* Though the magnitude of V_1 and V_2 is same, but directions are different.